



Predictive Demand Modeling for Inventory Optimization: Public, Anonymized Version

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Abstract— This document presents a public, anonymized summary of a demand forecasting solution developed in an academic–industrial datathon with a confidential partner. Using weekly aggregates from historical sales-order data, we conducted exploratory data analysis (EDA), identified temporal patterns, and evaluated predictive models (LSTM, SARIMA/ARIMA, Prophet) at the product level. The workflow covered data quality checks, seasonality and anomaly analysis, and per-product modeling strategies. Recurrent architectures performed robustly under diffuse seasonal signals, while SARIMA models proved effective for stable, well-structured series. This public version preserves structure and methodology while removing confidential identifiers; all figures and tables herein are generated from synthetic data.

Keywords—Demand forecasting, time series, LSTM, SARIMA, Prophet, seasonality, anomalies, inventory, anonymized public report.

PUBLIC VERSION & CONFIDENTIALITY NOTICE

This public version is intended for academic and portfolio use. The partner name is not disclosed. Any potentially sensitive elements (customer/product codes, operational policies, or raw internal figures) are omitted or generalized. All figures and optional tables are produced from synthetic data.

I. INTRODUCTION

This report summarizes a demand-forecasting approach to support inventory planning using weekly, product-level time series. The goal is to apply modern data analysis and predictive modeling to strengthen decision making in an industrial setting. This public version preserves methodology, detailed EDA structure, results, and conclusions while protecting confidentiality.

II. OBJECTIVE

We develop a product-level demand forecasting framework that:

- Aggregates sales to a weekly frequency per product and performs strict data quality controls.
- Characterizes series via seasonality, trend, and anomaly analysis to inform modeling choices.

- Trains and compares LSTM, SARIMA/ARIMA, and Prophet using standard error criteria.
- Produces an analytical report tailored for decision support in inventory planning.

III. DATA OVERVIEW (SYNTHETIC CONTEXT)

For this public version we provide non-sensitive context aligned with the original scope using synthetic assets:

- Time span: 2021–2023 (three years).
- Records: approximately 31,000 synthetic orders.
- Coverage: about 500 products and 900 clients.
- Units of sale: KG, L, PZA.

All visualizations and any optional tables below are generated from synthetic data and serve as illustrations of the methodology.

IV. METHODOLOGY

a. Exploratory Data Analysis

EDA focused on operational variables without exposing identifiers: order, creation date, client, item, quantity, and unit of sale. Core tasks:

- Data quality treatment of outliers and normalization of units.

- Weekly per-product resampling with explicit handling of missing weeks.
- Annual, monthly, and day-of-month summaries to uncover seasonal peaks and purchasing cycles.

1. Order ID

Unique order identifiers represent transactional groupings (client, product, quantities, timestamp). Multiple line items may share an order ID; aggregation enables downstream analysis at order or product-week level.

2. Order Creation Date

The order timestamp anchors temporal analysis. Weekly resampling was applied for modeling and yearly summaries were used to detect seasonality and specific peaks.

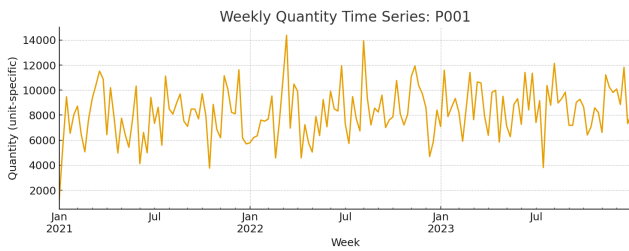


Fig. 1: Illustrative weekly time series for a top product (synthetic).

3. Client Code

Client identifiers are anonymized. Demand concentration and client activity distribution were examined using synthetic counts.

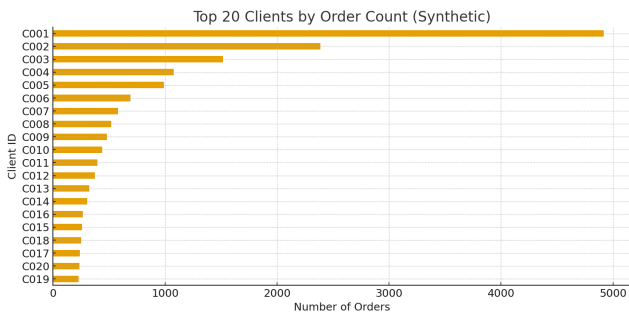


Fig. 2: Top 20 clients by order count (synthetic data).

4. Item (Product)

Product identifiers are anonymized. Product mix, rotation, and demand concentration were evaluated.

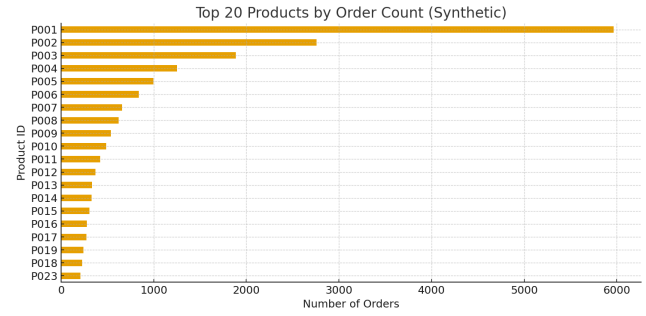


Fig. 3: Top 20 products by order count (synthetic data).

5. Quantity

Quantities were normalized per unit of sale (KG/L with decimal precision, PZA as integer counts). Extreme spikes were handled via winsorization or model-robustness checks.

6. Unit of Sale

Units (KG, L, PZA) inform scaling and interpretability. Unit-specific behavior is inspected in EDA and respected in modeling and denormalization.

b. Seasonality, Trend, and Anomalies

Classical decomposition (`seasonal_decompose`) and anomaly screening with *z-score* and *KMeans* segmentation of weeks were applied to identify seasonal windows, stable periods, and atypical weeks for treatment.

c. LSTM

- Preprocessing using `MinMax` scaling per product and embeddings to capture cross-series affinities.
- Sequencing with 12-week windows and one-step horizon.
- LSTM architecture combining temporal sequence and product embedding, followed by dense layers with `ReLU` activation and dropout.
- Training with the `adam` optimizer, MSE loss, and early stopping.

d. Prophet

Weekly series with logistic or linear growth models were used, incorporating weekly and annual seasonality. A short forecast horizon (four weeks) illustrated comparative performance.

e. SARIMA

Stationarity checks (ADF), regular and seasonal differencing, and model order selection using `auto_arima` were performed. Residual diagnostics (ACF/PACF) confirmed white-noise behavior and adequacy of fit.

V. ILLUSTRATIVE TABLES (SYNTHETIC DATA)

**TABLE 1:** TOP PRODUCTS BY ORDER COUNT (SYNTHETIC).

Rank	Product ID	Orders
1	P001	380
2	P002	275
3	P003	240
4	P004	230
5	P005	220

TABLE 2: TOP CLIENTS BY ORDER COUNT (SYNTHETIC).

Rank	Client ID	Orders
1	C001	420
2	C002	300
3	C003	270
4	C004	240
5	C005	200

VI. RESULTS

LSTM

The LSTM model performed stably on series with diffuse seasonality and non-linear dynamics, avoiding negative predictions and handling intermittent zeros effectively.

SARIMA

SARIMA provided the best fit for products with well-defined seasonality and stable demand, with narrow confidence intervals and satisfactory residual diagnostics.

Prophet

Prophet offered transparent component decomposition and interpretability, though it required careful tuning for irregular or sparse time series.

VII. CONCLUSIONS

A product-level forecasting framework combining EDA, temporal characterization, and multiple model families (LSTM, SARIMA/ARIMA, Prophet) was developed. Principal findings:

- Model performance is highly dependent on series morphology; prior classification improves accuracy.
- Criteria such as minimum active weeks, consistent seasonality, and temporal continuity are vital for stable training.
- Horizon-wise cross-validation and aggregate metrics (MAPE, MAE, RMSE) should guide model selection.
- Incorporation of non-sensitive exogenous factors (calendar effects, domain indicators) and reproducible automation enhance scalability.

Public Data and Figures Policy. All figures and tables in this public version were generated from synthetic or anonymized data. Product and client codes are placeholders and cannot be traced to real entities.